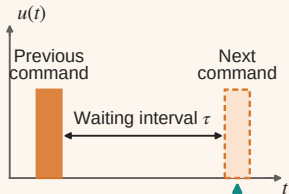


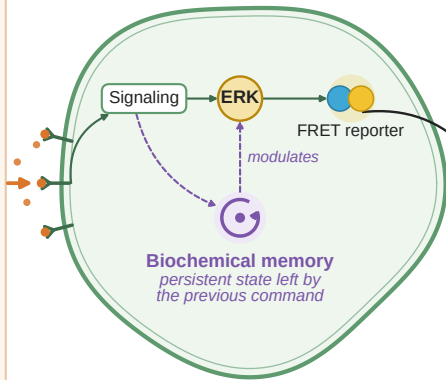
Repeated chemical commands



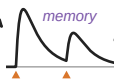
Chemical transmitter
FGF2 stimulation



Living cellular receiver (PC-12 cell)

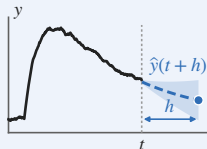


Observed
response $y(t)$



1 Future-response prediction

Can we predict the future response?



Which information helps?

Current state

Temporal history

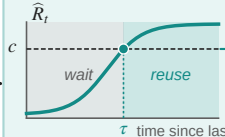
Mechanistic information

Evidence: Single-cell experiments

Mechanistic model

2 Adaptive receiver reuse

When can the receiver be reused?



$$\hat{R}_t \geq c$$

⇒ issue next command

Evidence: Mechanistic model (simulated receiver)

Adaptive reuse: issue the next command once $\hat{R}_t \geq c$